

PAUL GERHART

mail@paul-gerhart.de | paul-gerhart.de | Privacy Enhancing Technologies, TU Wien

EDUCATION

09/2024 - present	Doctoral Student in Cryptography, TU Wien Threshold Signatures, PoPRFs, Authentication Protocols (advisor: Dominique Schröder)
08/2022 - 08/2024	Doctoral Student in Cryptography, FAU Erlangen-Nürnberg Adaptor Signatures, Threshold Signatures (advisor: Dominique Schröder)
10/2020 - 07/2022	Master of Science in Computer Science, FAU Erlangen-Nürnberg Cryptography, Privacy, Cyber Security, and IT-Forensics (advisor: Dominique Schröder)
10/2017 - 09/2020	Bachelor of Science in Mathematics, University of Bonn Algebraic Topology (advisor: Wolfgang Lück)

EXPERIENCE

09/2024 - present	University Research Assistant, TU Wien Privacy Enhancing Technologies
12/2022 - present	Consultant and Software Developer, Freelance Cryptography, Privacy, IT-Forensics
08/2022 - 08/2024	University Research Assistant, FAU Erlangen-Nürnberg Applied Cryptography
10/2017 - 07/2022	Student Employee, Richard Wenzel GmbH & Co.KG Software Development

RESEARCH VISITS

06/2026 - 08/2026	Research Internship, Prof. Emma Dauterman Stanford University, Stanford, CA
08/2025 - 10/2025	Research Internship, Prof. Thomas Ristenpart Cornell Tech, New York, NY
08/2024	Research Visit, Prof. Anna Lysyanskaya Brown University, Providence, RI
08/2023 - 11/2023	Research Internship, Prof. Jonathan Katz University of Maryland, College Park, MD

HONORS & AWARDS

2025	Google PhD Fellowship for Privacy, Safety, and Security Google Research
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ADVISED STUDENTS

2025	Jay Taylor (Bachelor Thesis) <i>Delegated Fair Exchange via Threshold Wallets</i>
2024	Fabian Jost (Master Thesis) <i>Foundations of Blind Conditional Signatures</i>
2023	Daniel Lang (Bachelor Thesis) <i>Strong Unlinkability for Adaptor Signatures</i>
2023	Ruben Baecker (Master Thesis) <i>Phoenix Rise Once Again: How to Defeat the (PW-)Hero</i>
2023	Julian Thomas (Master Thesis) <i>Signature Security Revisited: One Dice to Rule Them All</i>

TEACHING

Winter 25/26	Privacy Enhancing Technologies (B.Sc.) Teaching assistant
Summer 25	Advanced Privacy Enhancing Technologies (M.Sc.) Teaching assistant
Winter 24/25	Privacy Enhancing Technologies (B.Sc.) Teaching assistant
Winter 23/24	Introduction to Modern Cryptography Teaching assistant
Winter 22/23	Introduction to Modern Cryptography Teaching assistant

COMMUNITY SERVICE

Program Committees

USENIX Security	2027
European Symposium on Research in Computer Security (ESORICS)	2026
SECURITY	2025, 2026
Crypto Valley Conference (CVC)	2025

External Reviews (Selected)

Advances in Cryptology (CRYPTO)	2023–2026
Advances in Cryptology (EUROCRYPT)	2023, 2024
Advances in Cryptology (ASIACRYPT)	2022
ACM Conference on Computer and Communications Security (CCS)	2024–2026
Public Key Cryptography (PKC)	2023–2025
European Symposium on Research in Computer Security (ESORICS)	2022–2026
IEEE Computer Security Foundations Symposium (CSF)	2023
Financial Cryptography and Data Security (FC)	2024

PUBLICATIONS

13 peer-reviewed publications

- 8 publications at top cryptography conferences (CRYPTO, EUROCRYPT, ASIACRYPT)
- 2 publications at top security conferences (ACM CCS, NDSS)
- 2 publications at top privacy venues (PoPETS)

BitPriv: A Privacy-Preserving Protocol for DeFi Applications on Bitcoin Ioannis Alexopoulos, Zeta Avarikioti, Paul Gerhart, Matteo Maffei, and Dominique Schröder	NDSS 2027
Fully Adaptive FROST with Identifiable Aborts from AOMDL Ruben Baecker, Paul Gerhart, Davide Li Calsi, Luigi Russo, Dominique Schröder, and Arkady Yerukhimovich	ASIACRYPT 2026
Fair Distributed Exchange via Threshold Adaptor Signatures Ruben Baecker, Paul Gerhart, Jonathan Katz, and Dominique Schröder	ASIACRYPT 2026
Universally Composable Adaptor Signatures Paul Gerhart, Daniel Rausch, and Dominique Schröder	ACM CCS 2026
Fully-Adaptive Two-Round Threshold Schnorr Signatures from DDH Paul Gerhart, Davide Li Calsi, Luigi Russo, and Dominique Schröder	EUROCRYPT 2026
How To Make Delegated Payments on Bitcoin: A Question for the AI Agentic Future Jay Taylor, Paul Gerhart, Sri Aravinda Krishnan Thyagarajan	FC 2026
Password-Hardened Encryption Revisited Ruben Baecker, Paul Gerhart, and Dominique Schröder	ASIACRYPT 2025

Universally Composable Password-Hardened Encryption	ASIACRYPT 2025
Behzad Abdolmaleki, Ruben Baecker, Paul Gerhart, Mike Graf, Mojtaba Khalili, Daniel Rausch, and Dominique Schröder	
A Fully-Adaptive Threshold Partially-Oblivious PRF	CRYPTO 2025
Ruben Baecker, Paul Gerhart, Daniel Rausch, and Dominique Schröder	
SoK: Descriptive Statistics Under Local Differential Privacy	PoPETS 2025
René Raab, Pascal Berrang, Paul Gerhart, and Dominique Schröder	
Measuring Conditional Anonymity - A Global Study	PoPETS 2024
Pascal Berrang, Paul Gerhart, and Dominique Schröder	
Foundations of Adaptor Signatures	EUROCRYPT 2024
Paul Gerhart, Dominique Schröder, Pratik Soni, and Sri Aravinda Krishnan Thyagarajan	
Practical Schnorr Threshold Signatures without the Algebraic Group Model	CRYPTO 2023
Hien Chu, Paul Gerhart, Tim Ruffing, and Dominique Schröder	
A General Approach to Adaptor Signatures	Master Thesis
advised by Dominique Schröder, Sri Aravinda Krishnan Thyagarajan, and Pratik Soni	
Surgery and a Theorem of Lickorish	Bachelor Thesis
advised by Wolfgang Lück and Christoph Winges	